SU TUNG LOC

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SKILLS

HPC:	Slurm, Spack, Warewulf, Apptainer, Docker, Linux, HPL and HPCG benchmarking.
DRL methods:	Reinforce, PPO, Q-learning, DQN, SARSA, A2C.
Machine Learning:	Python, PyTorch, TensorFlow, Scikit-learn, YOLO, RT-DETR, OpenCV, Gymnasium.
AI:	LangChain, LLMs, RAG, Prompt Engineering.
Parallel computing:	C, MPI, OpenMP, CUDA.
Cybersecurity & Networking:	Wireshark, Nmap, netcat, Burp Suite, Ghidra
SW/Web Development and GIS:	Java, JavaScript, Git, VS Code, SQL, NoSQL, HTML, CSS, Bootstrap, Leaflet, D3, DC.js
Others:	AWS, Weights & Biases (wandb), Anyscale, Foxglove, RViz2, Glim.
Languages	English, Vietnamese, Chinese

EDUCATION

Master of Science in Computer Science — Expected: May 2026
University of New Mexico — GPA: 4.18/4.0

Bachelor of Science in Computer Science — May 2024
University of New Mexico — *Magna Cum Laude* GPA: 3.88/4.0

Associate Degree — Aug 2019
Peninsula College — *Magna Cum Laude* GPA: 3.92/4.0

PROJECTS

HPL benchmarking | *Slurm, Bash, Git, OpenMPI, OpenBLAS*
- Performed HPL benchmarking on a multi-node HPC cluster using Slurm, OpenMPI, optimizing build parameters along with MPI communication to measure peak FLOPS performance, and evaluate the cluster architecture behavior.

HPCG benchmarking | *Slurm, Apptainer, Git, OpenMPI, NVIDIA container*
- Performed HPCG benchmarking on both CPU and GPU configurations using MPI and Slurm to evaluate memory bandwidth, communication overhead, and real-world solver performance.

Lunar Landar Training Analysis | *Python, PyTorch, OpenAI, Gymnasium, Matplotlib*
- Implemented and compared REINFORCE, SARSA, and DQN algorithms for Lunar Lander using PyTorch and Gymnasium, analyzing how exploration strategies and hyper-parameters impact learning performance.

Reinforcement Learning for Motion Planning and the Control of Dynamical Systems | *Python, PyTorch*
- Developed a 2D motion planning and vehicle control system using Q-learning, creating a custom RL environment and agent for automatic parking and optimal path navigation based on ODE-based simulations.

Driving Assistant | *Python, PyTorch, OpenCV, HuggingFace, YOLO*
- Built a Python-based dash-cam analysis system that integrates object detection, LLM-generated commentary, and automated pipelines to detect road objects, produce annotated video outputs, and generate contextual driving alerts.

Implement PPO on Atari Pong | *Python, PyTorch, OpenCV, Numpy, OpenAI, Gymnasium, Matplotlib*
- A state-of-the-art deep reinforcement learning algorithm that train an agent in playing Pong game

NYC Tree Census Web Map | HTML/CSS/JavaScript, Leaflet, DC, D3, Crossfilter, Bootstrap
- Developed an interactive web map visualizing the 2015 NYC Tree Census, enabling exploration of tree counts, species, and health status across all five boroughs through dynamic charts and filters.

Hexapawn | Java, JavaFX
- Developed a Hexapwn game in Java utilizing heuristic learning approach to create an adaptive unbeatable computer opponent.

Modulo time table visualization | Java, JavaFX
- Developed an interactive data visualization tool to map modular arithmetic multiplication tables, translating abstract number theory into dynamic geometric patterns.

Live Auction Distribution | Java, JavaFX, Socket, Network
- Developed an interactive web map visualizing the 2015 NYC Tree Census, enabling exploration of tree counts, species, and health status across all five boroughs through dynamic charts and filters.

Research

Countering of Block Withholding for Miners |

- Developed a theoretical model and simulation to analyze Bitcoin Block Withholding attacks, demonstrating that honest miners can protect their profitability by optimally allocating computational resources between pool and solo mining.

Analysis of Distributed Denial of Service Attacks: An Internal Perspective |

- Conducted a theoretical and empirical analysis of Distributed Denial of Service attacks by simulating attack scenarios using Docker containers and testing against a live Cloudflare-protected website, demonstrating that high concurrency requests exponentially degrade server performance beyond a critical threshold

Analysis of Censorship Evasion Tools |

- Evaluated the performance, privacy, and obfuscation capabilities of NordVPN, Tor, and Psiphon through empirical testing and case studies across high-censorship regions.

EXPERIENCE

UNM x Multiply Lab — Albuquerque, NM
AI Research Assistant — May 2026 – Present

- Conducted research to develop robotic systems that enable the manufacturing of advanced, next-generation therapies at scale and an affordable cost.

University of New Mexico — Albuquerque, NM
Teaching Assistant — Aug 2025 – June 2026

- Assisted faculty with course instruction, project guidance, and development of teaching materials.
- Held student consultations for homework and graded exams, quizzes, assignments, and papers.

University of New Mexico — Albuquerque, NM
TechDen Sales Assistant — June 2025 – Aug 2025

- Managed and updated product data using internal databases and spreadsheets, ensuring accurate inventory records.
- Assisted with website updates, including product listings and content changes, to maintain current and accurate information.

University of New Mexico — Albuquerque, NM
Grader CS105/CS341L — Aug 2024 – May 2025

- Implemented new assignments. Reviewed and graded homework assignments according to established standards, and assisted students with questions to clarify concepts.

Explora x UNM CHTM — Albuquerque, NM
Student Engineering Intern — May 2024 – Aug 2024

- Assisted in projects involving advanced laser and microwave technologies while supporting STEM outreach initiatives for future generations.
- Performed clean room operations and conducted experiments using Electron Microscopy, Diode Analysis, XRD, and Photoluminescence spectroscopy.